📘 **AI/ML Internship – Day 7 Notes & Tasks**

🟢 **Module 3: Python for AI (Revision + Assessment Day)**

📌 **Part 1: Quick Revision (Important Concepts)**

🔹 **Python Basics Recap**

- Variables & Data Types
- Input / Output
- Operators

🔹 **Control Statements**

- If-Else
- Nested conditions

🔹 **Loops**

- For loop
- While loop

🔹 **Data Structures**

- List
- Dictionary
- Set

🔹 **Functions**

- Function definition
- Parameters & return

🔹 **File Handling**

- Read / Write / Append

🔹 **Exception Handling**

- Try / Except / Finally

📌 **Part 2: Coding Patterns Revision**

🔹 **Pattern Example**

```
for i in range(1, 5):
    print("*" * i)
```

🔹 **Number Pattern**

```
for i in range(1, 6):
    for j in range(1, i+1):
        print(j, end=" ")
    print()
```

📌 **Part 3: Assessment (Important)**

🧠 **Section A: Theory (Short Answers)**

1. What is Python?
2. What is list and dictionary?
3. Difference between for loop and while loop
4. What is function?
5. What is exception handling?

💻 **Section B: Coding Test**

Q1:

Write program to check palindrome

Q2:

Find largest number in list

Q3:

Count vowels in string

Q4:

Create dictionary and print values

🚀 **Section C: Mini Task**

👉 Create program:

- Take 5 student inputs
- Store in list
- Find average
- Find topper

🎯 **Tasks for Day 7**

✅ **Task 1: Practice Problems**

1. Reverse list
2. Count words in string
3. Remove duplicates
4. Find second largest